Higher National Diploma in Engineering (Civil/Mechanical),

2nd Year, Second Semester Examination 2019

ME 2217 Mechanics of Solids.

Question 1.

Consider a rectangular plate (Fig. 01) with sides a and b of thickness t.

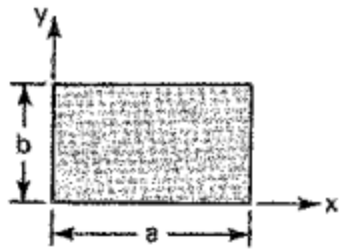

Fig. 01

a) Show that the stress function $\emptyset = -\frac{p}{b^a} xy^2(3b - 2y)$is a valid function5. (08 marks)

b) Determine the stresses $\sigma x, \sigma y$ and for the above stress function where p is a constant. (04 marks)

c) Find the resultant normal and shearing boundary forces along all edges of the plate. (08 marks)

Question 2.

a) State Castigliano's Second theorem8. (05 marks)

b) Consider the cantilever beam with a load F at a distance b from the free end (Fig. 02)9. Find the deflection at the free end10. (15 marks)

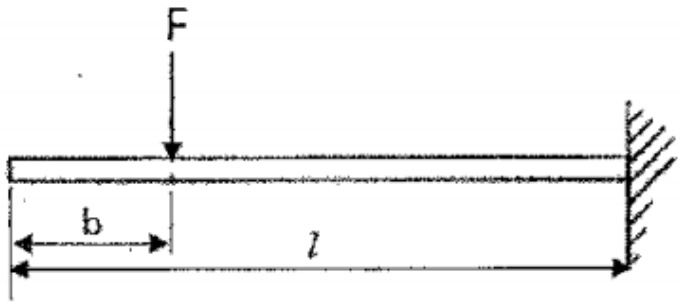

Fig. 02

Question 3.

Structural steel channels with E=200 GPa are used for a **12 m column** as shown in Fig. 03. Section properties of a single channel (Fig.03.a) is given below:

- Area (A) =3795mm^2
- I_{x1x1}=25.3x10^6 mm^4
- I_{y1y1}=1.01x10^6 mm^4

Determine the **total compressive load required to buckle the column** (**both ends hinged**) if:

a) One channel is used as in Fig. 03.a. (10 marks)

b) Two channels are placed 150 mm back to back as shown in Fig. 03.. (10 marks)

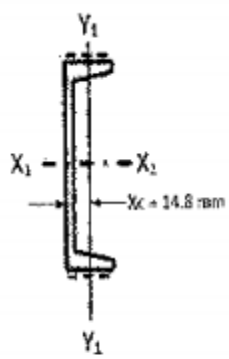

Fig.03.a

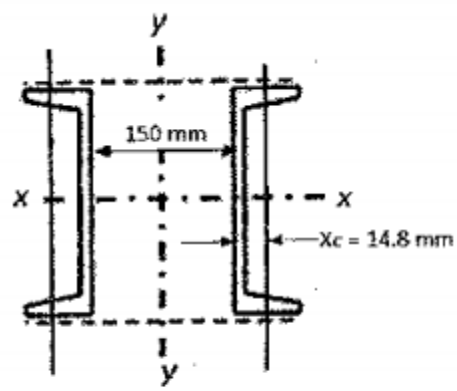

Fig.03.b

Question 4.

Fig.04 shows the **elliptical cross section** of a prismatic bar subjected to a **twisting moment**.

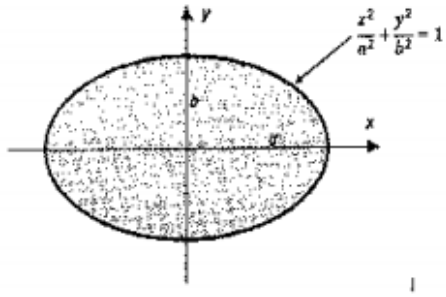

Fig.04

a) Obtain a suitable stress function. (04 marks)

b) Determine the angle of twist per unit length (θ). (04 marks)

c) Derive the shear stress distribution of the bar. (04 marks)

d) Find the maximum value of shear stresses. (04 marks)

e) Derive an expression for warping function w(x,y). (04 marks)

Question 5.

An arrangement of **statically indeterminate bars** is shown in fig 05. Node A and Node C are **fixed nodes** and cross-sectional areas of bar AB and BC are 1×10^{-4} m ^{2} and 2×10^{-4} m^2 respectively.If **10 kN** is applied at node B as shown in Fig. 05:

a) Displacement at point B. (10 marks)

b) Reaction forces at A and C (10 marks)

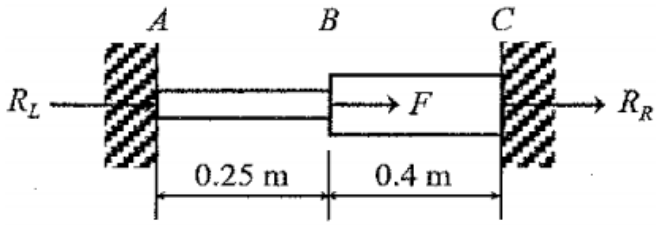

Fig. 5

Question 6.

a) What is meant by "failure of a structure or machine" in general terms? (03 marks)

b) Name four important variables connected with structural failure. (05 marks)

c) What is meant by "fracture" in material failure? (02 marks)

d) i) What are the main causes of fatigue failure? (02 marks)

ii) What do you measure in a simple fatigue test? (02 marks)

e) S-N diagram for steel is shown in fig 06. Briefly explain it. (06 marks)

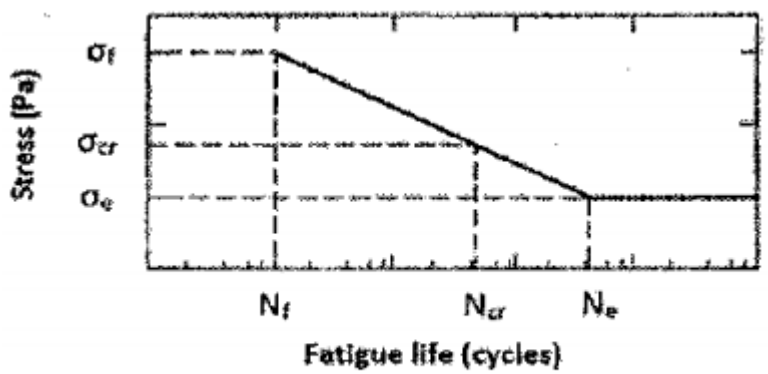

Fig.6